

Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: HAL LEAD-FREE
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE85,000 × 100,000 mm

BOARD THICKNESS1,600 mm

TRACE WIDTH0,300 mm

TRACE TO TRACE0,020 mm

MIN. HOLE (PTH)0,250 mm

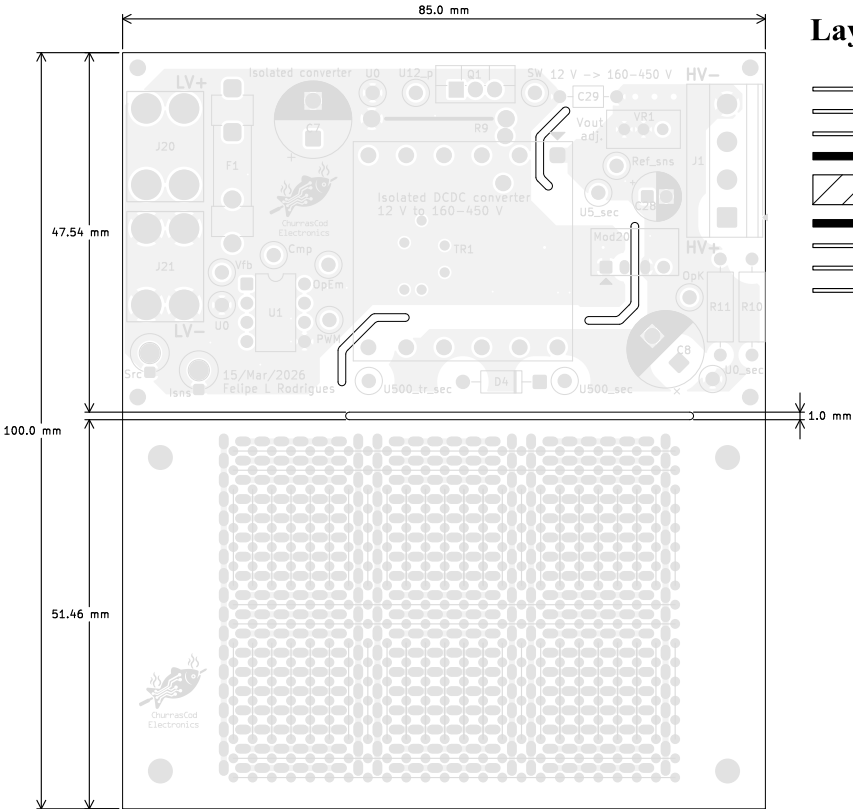
MIN. HOLE (NPTH)2,100 mm

ANNULAR RING0,150 mm

COPPER TO HOLE0,200 mm

COPPER TO EDGE0,250 mm

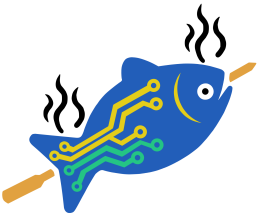
HOLE TO HOLE0,254 mm

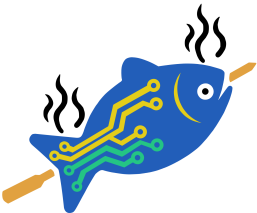


Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen			Direct Printing	Legend	GBR
F.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
Core		1,49 mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0,035 mm (1 oz)		Signal	GBR
B.Mask		0,02 mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

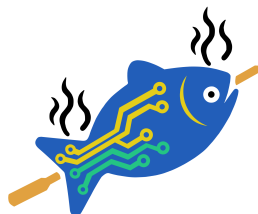
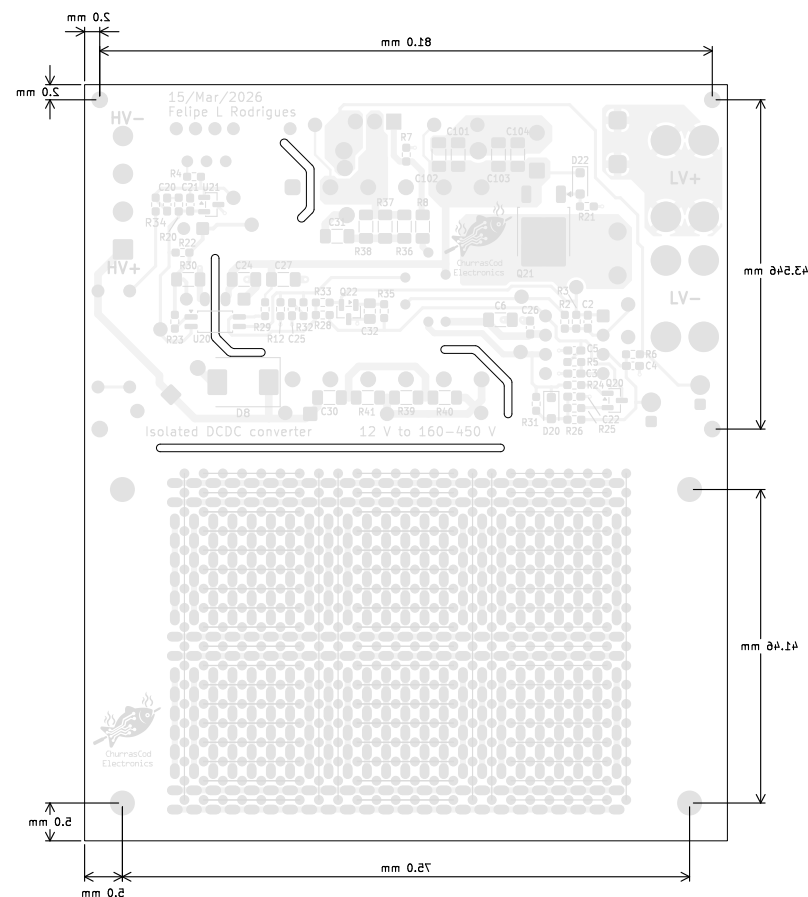
Total thickness: 1,6 mm
Note: external layer thicknesses are specified after plating



	Comments:		Company: NA		Variant: CHECKED	Git Hash: 26c8ce0
	Sheet Title: Top Fabrication (Scale 1:1)		Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	Sheet Path:		File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
			Reviewer: FR	Size: A4	Sheet: 1 of 7	

Step-Up Fabrication Document

Bottom Fabrication (Scale 1:1)



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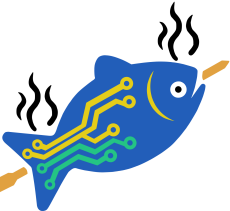
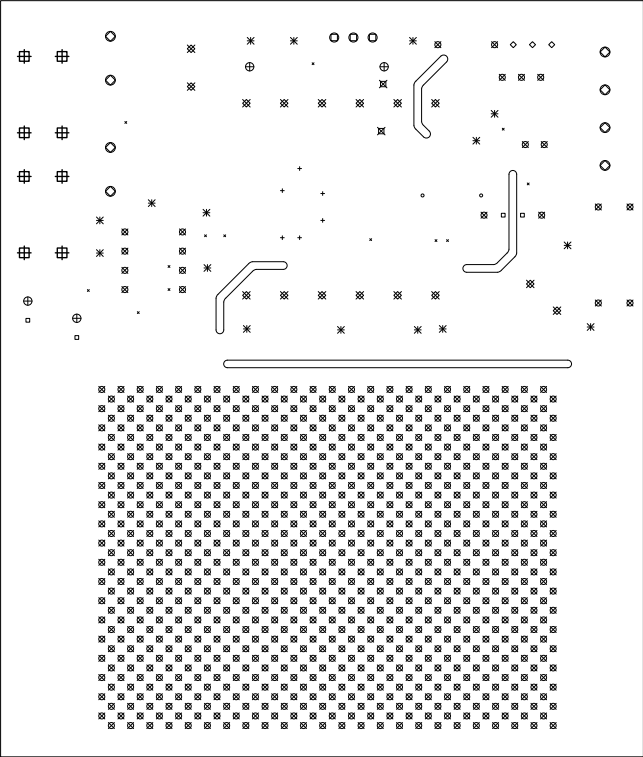
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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 2 of 7

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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0.40mm (15,75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0.50mm (19,69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0.70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	885	0.80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✱	16	0.90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊗	2	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⊗	16	1.00mm (39,37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1.10mm (43,31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	3	1.20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
○	8	1.30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	8	1.80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
Total 970						



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Comments:

Company:

NA

Variant:

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Git Hash:

26c8ee0

Board Name:

Step-Up

Project Name:

Isolated step-up 12:450 V

Sheet Title:

Drill Drawing (L1 - L4)

File Name:

Isolated_12to450V.kicad_pcb

Designer:

FR

Date:

2026-02-07

Revision:

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Sheet Path:

Reviewer:

FR

Size:

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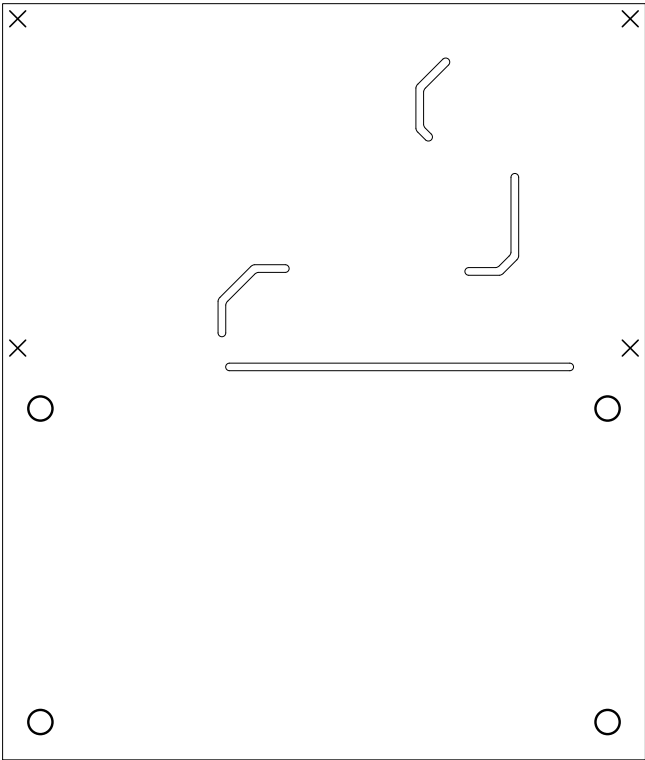
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Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2,10mm (82,68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
O	4	3,20mm (125,98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
Total 8						



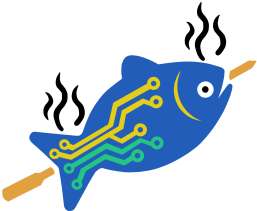
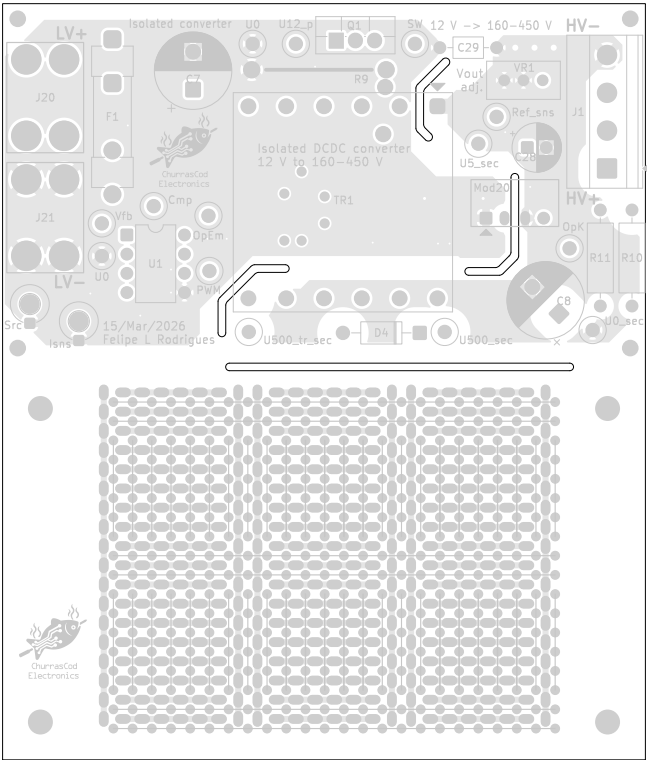
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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V		
Sheet Title: Drill Drawing (L1 - L4)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)	
Sheet Path:		Reviewer: FR	Size: A4	Sheet: 4 of 7	

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Top Test Points (Scale 1:1)

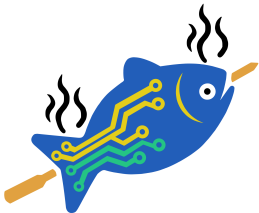
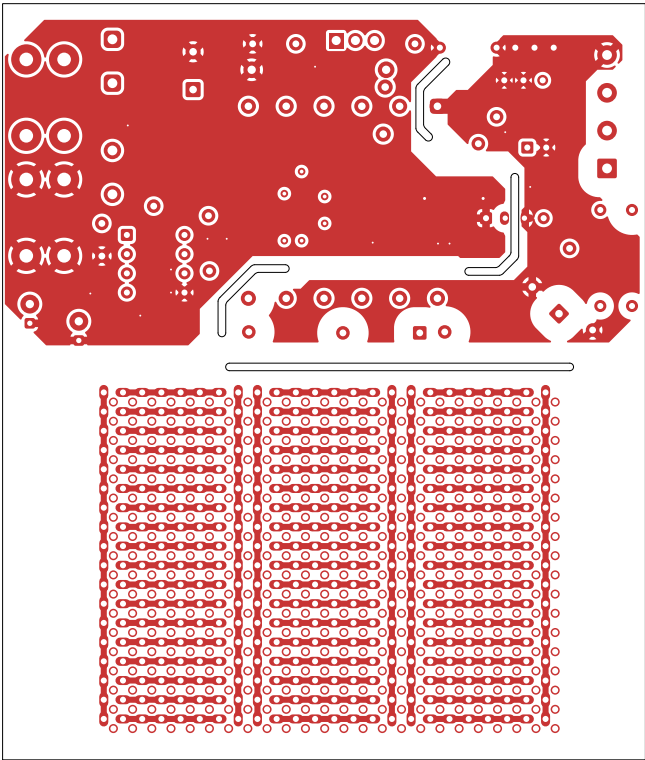
Ref.	Net	X [mm]	Y [mm]
TP9	Sec_ref_sns	69.73	88.65
TP10	Sec_OpK	79.38	71.25
TP20	SW	58.93	98.30
TP23	PWM	31.76	68.26
TP24	Vfb	17.53	74.55
TP25	Comp	24.39	76.83
TP26	OpEm	31.63	75.56
TP27	U500_sec	62.87	60.20
TP28	U0_sec	82.43	60.45
TP30	U500_tr_sec	36.96	60.20
TP31	U12_p	43.19	98.30
TP32	U0	17.53	70.23
TP33	U0	37.47	98.30
TP34	U5_sec	67.32	85.09
TP35	Isns,U0	14.48	60.34
TP36	Src,U0	8.01	62.61
TP101	Net-(TP101-Pad1)	20.33	52.20
TP102	Net-(TP101-Pad1)	22.87	52.20
TP103	Net-(TP101-Pad1)	25.41	52.20
TP104	Net-(TP101-Pad1)	27.95	52.20
TP105	Net-(TP101-Pad1)	30.49	52.20
TP106	Net-(TP101-Pad1)	33.03	52.20
TP107	Net-(TP107-Pad1)	20.33	49.66
TP108	Net-(TP107-Pad1)	22.87	49.66
TP109	Net-(TP107-Pad1)	25.41	49.66
TP110	Net-(TP107-Pad1)	27.95	49.66
TP111	Net-(TP107-Pad1)	30.49	49.66
TP112	Net-(TP107-Pad1)	33.03	49.66
TP113	Net-(TP113-Pad1)	20.33	47.12
TP114	Net-(TP113-Pad1)	22.87	47.12
TP115	Net-(TP113-Pad1)	25.41	47.12
TP116	Net-(TP113-Pad1)	27.95	47.12

All dimensions are in millimeters unless otherwise specified.



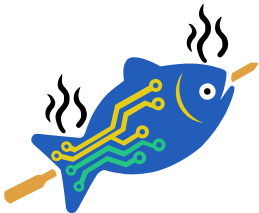
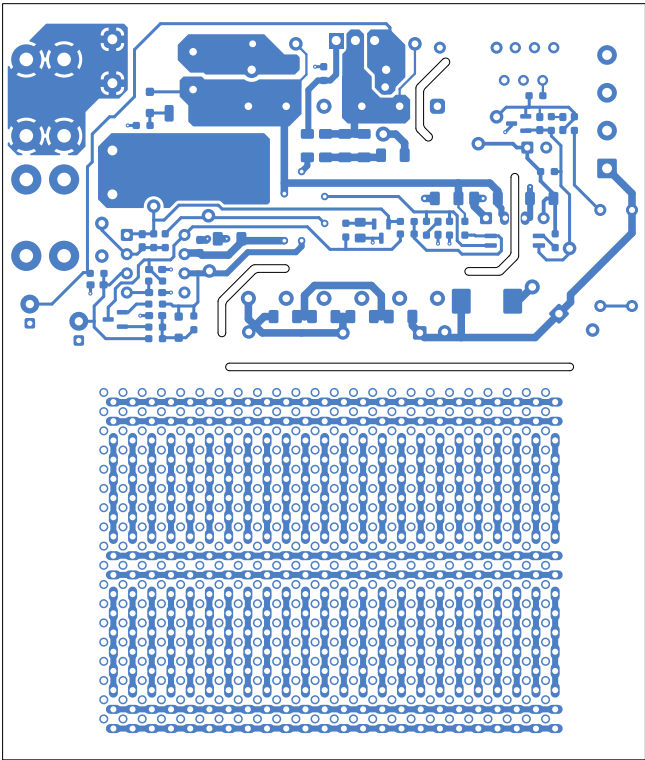
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	Board Name: Step-Up		Project Name: Isolated step-up 12:450 V	
	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
	Reviewer: FR		Size: A4	Sheet: 5 of 7



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F.Cu (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2026-02-07	2.0.0+ (Unreleased)
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Reviewer: FR		Size: A4		Sheet: 7 of 7		